

CSE3267 DATA MINING - PROJECT REPORT

Titanic Survival Prediction using Decision Trees

Field	Information
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Abstract

This project investigates passenger survival prediction on the Titanic dataset using a Decision Tree classification algorithm. The main objective is to identify the most influential factors that affected survival and to build an interpretable classification model. The dataset was obtained from the Kaggle “Titanic: Machine Learning from Disaster” competition and contained 891 passenger records with demographic and travel-related features. Before training the model, missing values were handled, irrelevant columns were removed, and categorical variables were encoded into numerical format. The model was trained using an 80/20 train-test split and controlled with a maximum depth of 4 to reduce overfitting. The final Decision Tree model achieved a test accuracy of 79.88%. The results showed that gender, passenger class, fare, and age were the strongest predictors of survival. Female passengers and passengers from higher classes had significantly better survival chances. Overall, this project demonstrates that Decision Trees are useful not only for prediction, but also for explaining patterns in historical data through a transparent decision-making structure.

1. Introduction

The sinking of the RMS Titanic remains one of the most well-known maritime disasters in history. On April 15, 1912, the Titanic sank after colliding with an iceberg during its maiden voyage, leading to the deaths of many passengers and crew members. Although survival depended partly on chance, historical evidence and data analysis show that some groups of people had a higher probability of survival than others. In particular, women, children, and passengers from higher social classes were more likely to survive.

The Titanic dataset has become a classic benchmark in Data Mining and Machine Learning because it contains a clear prediction target and a mixture of numerical, categorical, and text-based variables. This makes it suitable for practicing the full data mining pipeline: data understanding, preprocessing, exploratory data analysis, feature selection, model training, evaluation, and interpretation.

The purpose of this project is to use a Decision Tree classifier to predict whether a passenger survived based on socio-demographic and travel-related features. The project also aims to interpret the model output and explain which variables influenced the predictions the most. This is important because a good data mining model should not only produce accurate predictions but also provide meaningful insights about the data.

- To clean and prepare the Titanic passenger dataset for machine learning.
- To explore survival patterns using exploratory data analysis.
- To train a Decision Tree classification model.
- To evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.
- To interpret the model using feature importance and decision tree visualization.

2. Literature Review

Numerous studies have explored machine learning techniques for predicting Titanic passenger survival. Exploratory Data Analysis is widely recognized as an important first step because it helps researchers understand the dataset before applying any algorithm. Kakde and Agrawal (2018) emphasized that EDA helps reveal relationships between variables and reported competitive Decision Tree accuracy. Ai (2023) also evaluated models such as Artificial Neural Networks and Support Vector Machines and showed that proper feature engineering and removal of unnecessary data can significantly improve performance.

Other comparative studies support the idea that preprocessing strongly affects model performance. Durmuş and İşçi Güneri (2020) analyzed the Titanic dataset using generalized linear models and Decision Trees and showed that data cleaning and removing variables with little contribution improved the classification rate. Ekinici et al. (2018) compared several data mining techniques and demonstrated that classification accuracy depends on both model choice and dataset characteristics.

More recent research has continued to use the Titanic dataset as a benchmark for testing modern machine learning algorithms. Shaik et al. (2024) explored machine learning algorithms for survival forecasting, while Sreenivasulu and Chandrasekar (2025) compared Logistic Regression and Decision Trees. Across these studies, gender, passenger class, and age repeatedly appear as important survival factors.

Based on the reviewed literature, Decision Trees are a suitable choice for this project because they provide a balance between prediction performance and interpretability. While more complex models may achieve higher accuracy, Decision Trees allow the decision logic to be visualized and explained clearly.

3. Dataset Description

The dataset used in this project is the train.csv file from the Kaggle “Titanic: Machine Learning from Disaster” competition. The raw dataset contains 891 passenger records and 12 columns. The target variable is Survived, where 0 means that the passenger did not survive and 1 means that the passenger survived. The remaining columns describe passenger identity, demographic characteristics, family relationships, ticket information, fare, cabin, and port of embarkation.

Feature	Description	Type
PassengerId	Unique identifier for each passenger	Numerical
Survived	Target variable: 0 = died, 1 = survived	Binary
Pclass	Ticket class: 1st, 2nd, or 3rd	Categorical / Numerical
Name	Passenger name	Text
Sex	Passenger gender	Categorical
Age	Passenger age	Numerical
SibSp	Number of siblings or spouses aboard	Numerical
Parch	Number of parents or children aboard	Numerical
Ticket	Ticket number	Text
Fare	Passenger fare	Numerical
Cabin	Cabin number	Text
Embarked	Port of embarkation: C, Q, or S	Categorical

This dataset is appropriate for classification because the target variable has two classes: survived and did not survive. It also contains different feature types, which makes preprocessing necessary before applying a machine learning algorithm.

4. Exploratory Data Analysis

Exploratory Data Analysis was conducted to better understand the structure of the dataset and identify possible relationships between passenger characteristics and survival. This step is important because it helps explain why certain variables later become important in the Decision Tree model.

4.1 Survival Distribution

The dataset contains more passengers who did not survive than passengers who survived. This means the target variable is slightly imbalanced. Accuracy is still useful, but it should be interpreted together with other metrics such as precision, recall, and F1-score.

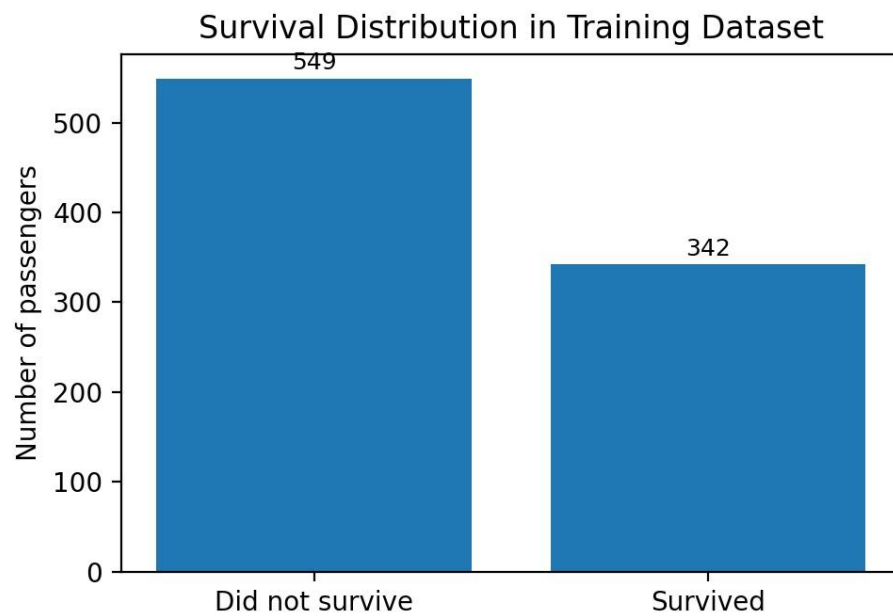


Figure 1. Survival distribution in the Titanic training dataset.

4.2 Survival by Gender

Gender had a strong relationship with survival. Female passengers survived at a much higher rate than male passengers. This pattern matches the historical evacuation principle commonly described as “women and children first.” This also explains why the Decision Tree selected Sex as the root split.

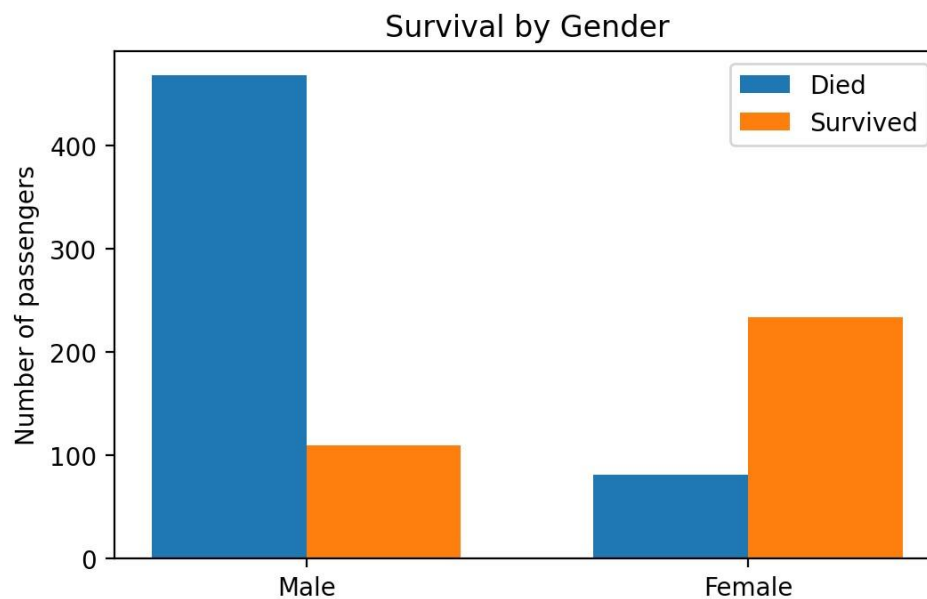


Figure 2. Survival comparison between male and female passengers.

4.3 Survival by Passenger Class

Passenger class also influenced survival. First-class passengers had the highest survival rate, while third-class passengers had the lowest. This suggests that socio-economic status and physical access to lifeboats may have affected survival opportunities.



Figure 3. Survival comparison by passenger class.

4.4 Age and Fare Considerations

Age and fare were also relevant variables. The Age distribution chart shows that most passengers were young adults, especially between 20 and 40 years old, while the Fare distribution is strongly right-skewed because most passengers paid low or moderate ticket prices and only a small group paid very high fares. These two graphs were added to make the exploratory analysis more complete before the model training stage. Since the Age column contained missing values, median imputation was used during preprocessing. Fare was kept as a numerical feature because it indirectly reflects passenger class and socio-economic status.

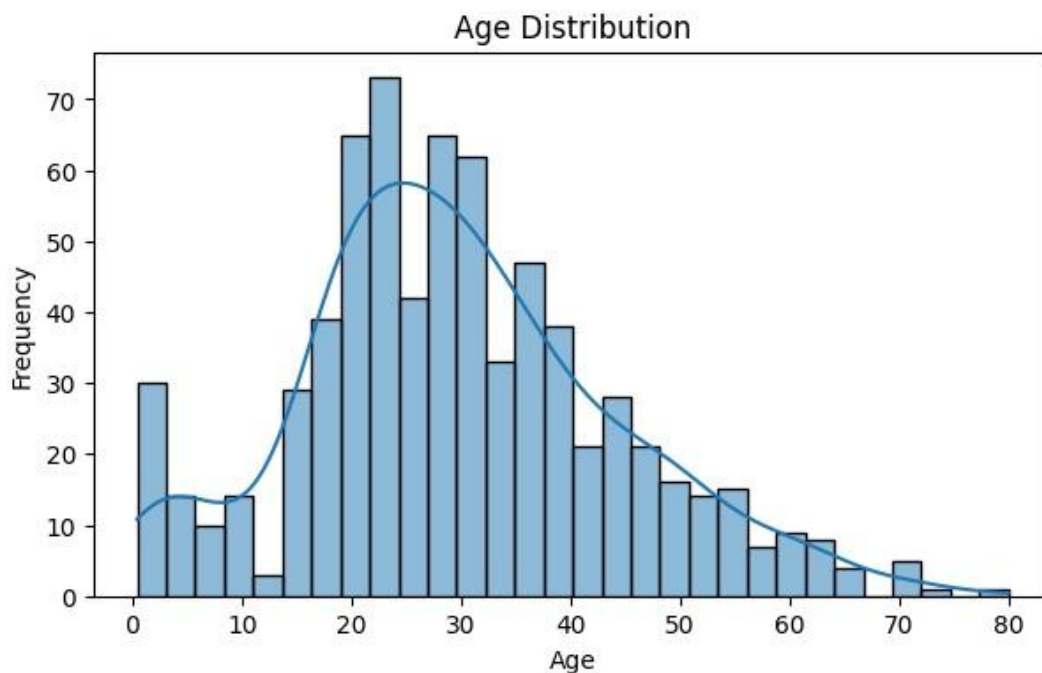


Figure 4. Age distribution of Titanic passengers.

The age distribution helps explain why Age was kept as one of the model features. Most passengers were adults, but there were also children and elderly passengers. Because many Age values were missing in the original dataset, replacing them with the median age provided a stable and simple preprocessing solution.

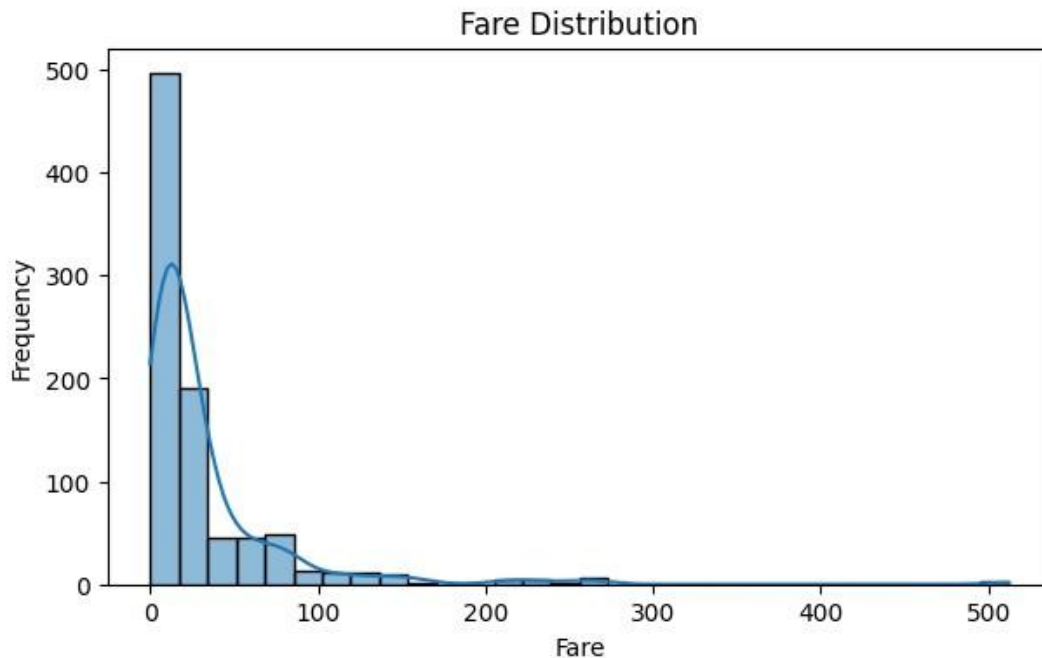


Figure 5. Fare distribution of Titanic passengers.

The fare distribution shows that the majority of passengers paid relatively low fares, while a small number paid much higher fares. This pattern is expected because ticket price was closely connected to passenger class. Therefore, Fare can provide useful information for the Decision Tree model.

5. Methodology

The project followed a systematic data mining pipeline implemented in Python. The main libraries used were pandas for data manipulation, matplotlib for visualization, and scikit-learn for machine learning. The workflow included data loading, data inspection, missing value handling, feature selection, categorical encoding, train/test splitting, model training, evaluation, and interpretation.

5.1 Data Preprocessing

Raw data usually contains missing values, text variables, and irrelevant columns that machine learning algorithms cannot process directly. Therefore, preprocessing was necessary before model training.

- Missing Age values were replaced with the median age. The median was chosen because it is less affected by outliers than the mean.
- Missing Embarked values were replaced with the most frequent embarkation port.
- The Cabin column was removed because it contained too many missing values, which could introduce noise.
- PassengerId, Name, and Ticket were removed because they do not directly help a basic Decision Tree model without additional feature engineering.
- The Sex column was encoded as male = 0 and female = 1.

- The Embarked column was encoded into numerical values so that it could be used by scikit-learn.

5.2 Feature Selection

The final selected features were Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked. These variables were selected because they represent passenger demographics, social status, family relationships, travel cost, and embarkation information. Removing unnecessary variables reduced dimensionality and helped the model focus on meaningful patterns.

5.3 Train-Test Split

The prepared data was divided into a training set and a testing set. The training set contained 80% of the records, while the testing set contained 20%. This approach allowed the model to learn from one part of the data and then be evaluated on unseen records. The test set contained 179 passengers.

6. Decision Tree Algorithm

A Decision Tree is a supervised learning algorithm used for classification and regression tasks. In classification, it predicts a class label by splitting the dataset into smaller subsets based on feature values. Each internal node represents a condition, each branch represents the result of that condition, and each leaf node represents a final prediction.

The algorithm chooses the best feature for each split by measuring how well the split separates the target classes. One common criterion is Gini Impurity. A lower Gini value means that a node contains records that mostly belong to one class. The Decision Tree tries to create splits that reduce impurity and make the child nodes more homogeneous.

Decision Trees are especially useful in this project because their logic can be interpreted visually. For example, if the first split is based on gender, the tree clearly shows that gender provides the strongest separation between survivors and non-survivors.

7. Model Training and Evaluation

After preprocessing, the DecisionTreeClassifier from scikit-learn was trained using the selected features. The maximum depth of the tree was limited to 4. This pruning strategy was used to reduce overfitting and keep the tree readable. Without limiting depth, a Decision Tree may memorize training data too closely and perform worse on unseen data.

The model was evaluated using several metrics. Accuracy measures the percentage of all correct predictions. Precision measures how many predicted survivors were actually survivors. Recall measures how many actual survivors were correctly identified. F1-score combines precision and recall into one balanced metric.

Metric	Formula / Meaning	Value
Accuracy	Correct predictions / all predictions	79.88%
Precision	$TP / (TP + FP)$	83.93%
Recall	$TP / (TP + FN)$	63.51%
F1-score	Harmonic mean of precision and recall	72.31%

The results show that the model was generally accurate. Precision was high, meaning that when the model predicted survival, it was usually correct. Recall was lower, meaning that the model missed some actual

survivors. This is important because false negatives represent passengers who survived but were predicted as not surviving.

8. Results and Discussion

8.1 Confusion Matrix Analysis

The confusion matrix gives a detailed view of the model’s prediction results on the test set. Out of 179 test records, the model correctly predicted 96 passengers as not surviving and 47 passengers as surviving. It incorrectly predicted 9 passengers as surviving when they did not survive, and 27 passengers as not surviving when they actually survived. This result supports the overall accuracy score but also shows that the model had more difficulty identifying all actual survivors.

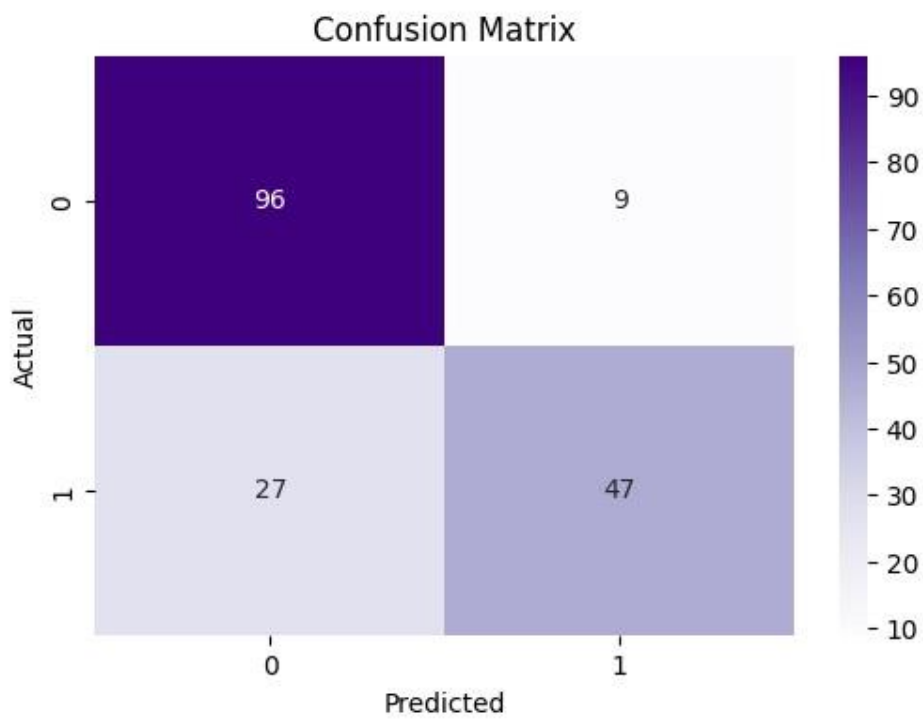


Figure 6. Confusion Matrix of the Decision Tree model.

Prediction Type	Count	Explanation
True Negative	96	Passenger did not survive and the model predicted did not survive.
True Positive	47	Passenger survived and the model predicted survived.
False Positive	9	Passenger did not survive but the model predicted survived.
False Negative	27	Passenger survived but the model predicted did not survive.

8.2 Feature Importance

Feature importance shows which variables contributed the most to the Decision Tree predictions. The most influential feature was Sex, followed by Pclass, Fare, and Age. Family relationship variables such as SibSp and Parch had a smaller effect in this model. The updated chart uses clearer axis labels so the importance scores can be interpreted more easily.

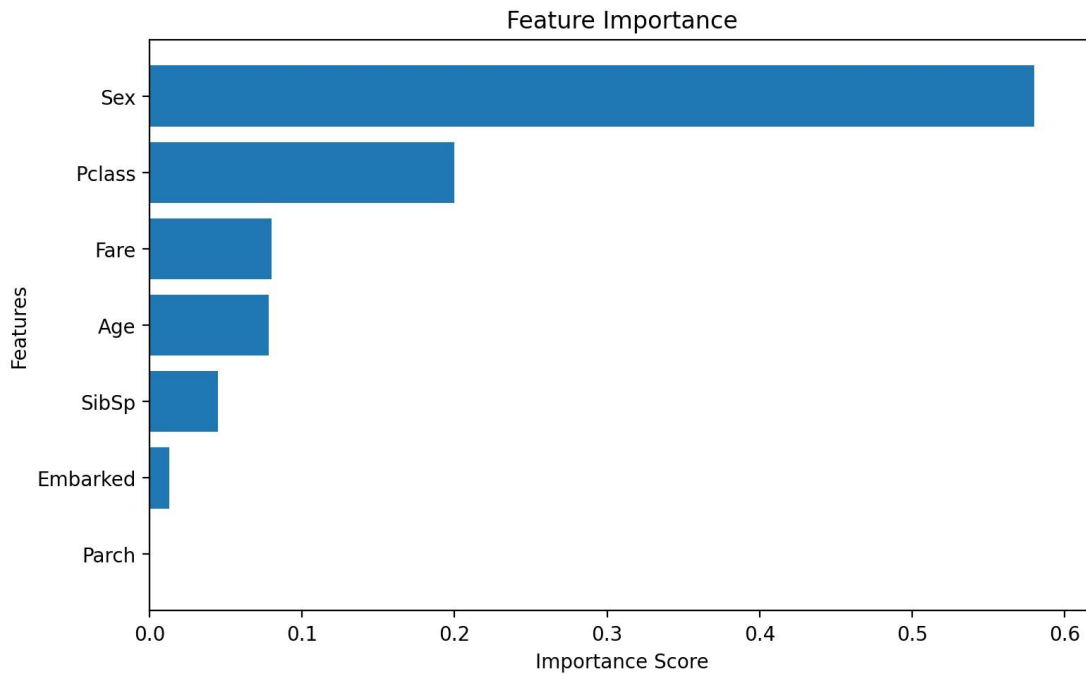


Figure 7. Feature Importance chart for the Decision Tree model.

8.3 Decision Tree Visualization

The Decision Tree visualization confirms the findings from the feature importance chart. The first major split is based on gender, which means that gender was the strongest predictor of survival. Female passengers, especially those in the first and second classes, were predicted to have a high probability of survival. Male passengers had a lower probability of survival, although young boys had better chances than adult males.

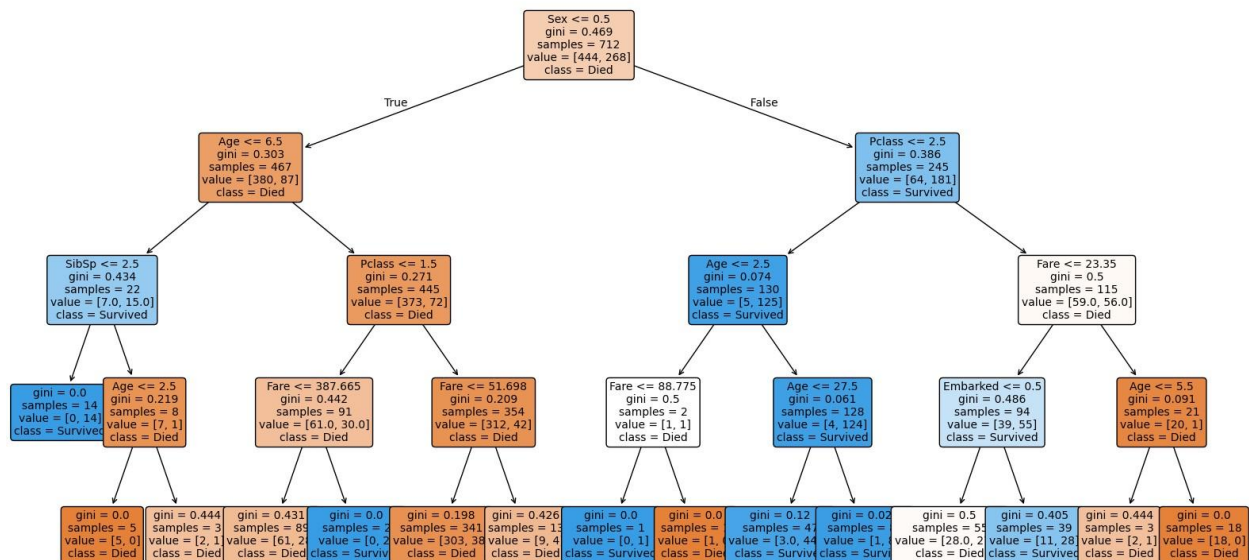


Figure 8. Decision Tree visualization showing the model decision logic.

8.4 Comparison with Other Models

Although this project focuses on Decision Trees, other classification algorithms could also be applied to the Titanic dataset. The table below compares possible alternatives.

Model	Advantage	Disadvantage
Logistic Regression	Simple, fast, and useful as a baseline	Less flexible for complex non-linear patterns
Decision Tree	Interpretable and easy to visualize	Can overfit if not pruned
Random Forest	Often improves accuracy by combining many trees	Less transparent than a single tree
Support Vector Machine	Can produce strong classification boundaries	Harder to interpret
Neural Network	Can model complex relationships	Requires more tuning and data

The Decision Tree was selected because the goal of this project was not only to predict survival but also to explain the prediction process. This makes it more suitable than black-box models for an academic data mining report.

9. Limitations

This project has several limitations. First, the Titanic training dataset contains only 891 records, which is relatively small for building highly generalizable machine learning models. Second, some variables had missing values. Although missing values were handled through imputation or column removal, this may still introduce uncertainty. Third, only basic feature engineering was used. More advanced features, such as passenger title extraction from names, could improve the model. Fourth, the Decision Tree model is sensitive to the train-test split, meaning slightly different results may occur if the data is divided differently. Finally, historical and social factors cannot be fully captured by the available dataset columns.

10. Future Work

Future work could improve the project in several ways. First, additional feature engineering could be applied. For example, the Name column could be used to extract titles such as Mr., Mrs., Miss., and Master. These titles may represent gender, age group, and social status more precisely. Second, a FamilySize feature could be created by combining SibSp and Parch, and an IsAlone feature could identify passengers who travelled alone.

Third, hyperparameter tuning with GridSearchCV could be used to find the best maximum depth, minimum samples per split, and splitting criterion. Fourth, cross-validation could provide a more stable evaluation than a single train-test split. Finally, ensemble models such as Random Forest and Gradient Boosting could be tested to improve accuracy, although they may reduce interpretability compared with a single Decision Tree.

- Extract passenger titles from the Name column.
- Create FamilySize and IsAlone features.
- Use GridSearchCV for hyperparameter optimization.
- Apply k-fold cross-validation.
- Compare Decision Tree with Random Forest, Gradient Boosting, Logistic Regression, and SVM.
- Use ROC-AUC and cross-validation scores for deeper evaluation.

11. Conclusion

In this project, a Decision Tree classifier was successfully applied to the Titanic dataset to predict passenger survival. The complete data mining process included dataset analysis, preprocessing, feature selection, model training, evaluation, and interpretation. The final model achieved an accuracy of 79.88%, which is competitive for a simple and interpretable model.

The analysis showed that gender and passenger class were the strongest survival factors. Female passengers and passengers from higher classes had a greater chance of survival, while male passengers, especially from lower classes, were less likely to survive. The Decision Tree visualization made these patterns clear and easy to understand. Therefore, the project demonstrates that Decision Trees are valuable not only for classification but also for explaining how data-driven decisions are made.

12. References

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